Support Vector Machines for Credit Scoring: a performance comparison between classic Machine Learning and Quantum-enhanced approach

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Quantum AI and risk management

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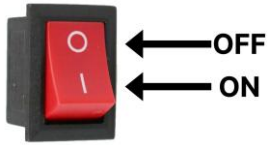
Why Credit scoring and QML?

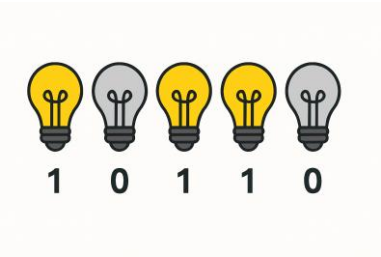
- Credit Scoring is a critical task in credit risk assessment
  - Credit and financial risk management
  - Counterparty's repayment capacity estimation through financial features
  - Financial solidity of the counterparty affects its repayment capacity
- To prevent credit losses, classification of clients in risk classes is fundamental
- Supervised ML models for classication as Support Vector Machines are useful for the division of clients in different risk classes to identify potential defaulting positions
- Quantum-enhanced Support Vector Machines may represent an innovative effective tool to make classification better than traditional models

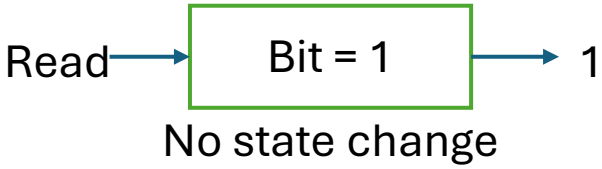
Fundamental elements

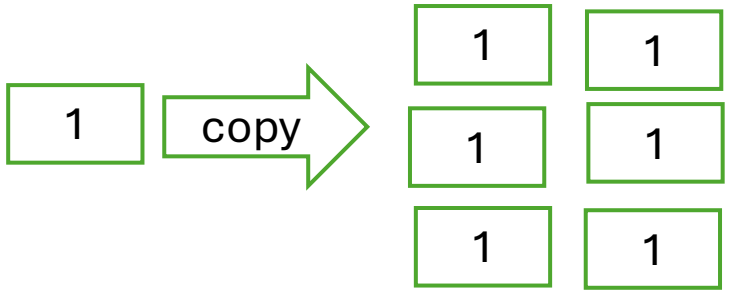
Classical bits

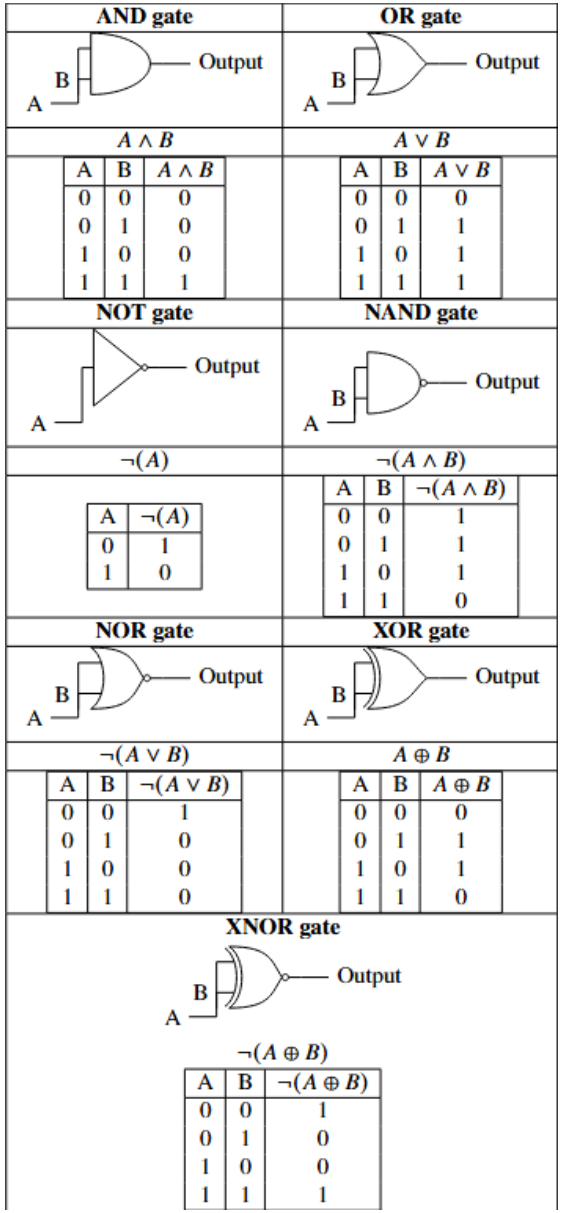
- **Binary state: 0 or 1**
- **Deterministic nature: no state alteration**
- **Local independence: no intrinsic correlation**
- **Unlimited replicability**
- **Logic gates representation: irreversibility**











AND gate

$A \wedge B$

A	B	$A \wedge B$
0	0	0
0	1	0
1	0	0
1	1	1

OR gate

$A \vee B$

A	B	$A \vee B$
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate

$\neg(A)$

A	$\neg(A)$
0	1
1	0

NAND gate

$\neg(A \wedge B)$

A	B	$\neg(A \wedge B)$
0	0	1
0	1	1
1	0	1
1	1	0

NOR gate

$\neg(A \vee B)$

A	B	$\neg(A \vee B)$
0	0	1
0	1	0
1	0	0
1	1	0

XOR gate

$A \oplus B$

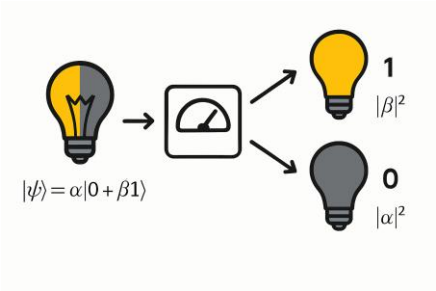
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

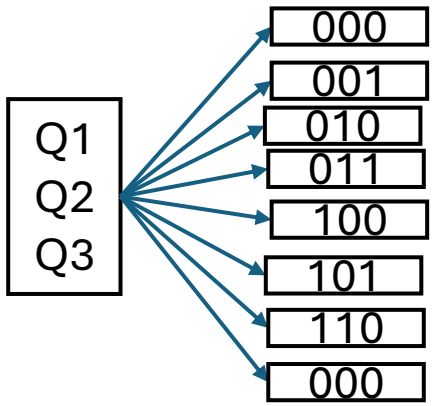
XNOR gate

$\neg(A \oplus B)$

A	B	$\neg(A \oplus B)$
0	0	1
0	1	0
1	0	0
1	1	1

Quantum bits





- **Superposition:**

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

  with $\alpha, \beta \in \mathbb{C}$
- **Probabilistic measurement $P(0) = |\alpha|^2, P(1) = |\beta|^2$ with $|\alpha|^2 + |\beta|^2 = 1$**
- **More information representable**

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle$$

- **Bloch sphere geometrical representation**

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle$$

- **Non-local correlation and entanglement**
- **No replicability: no-cloning theorem**

If $\theta = 0 \implies |0\rangle$

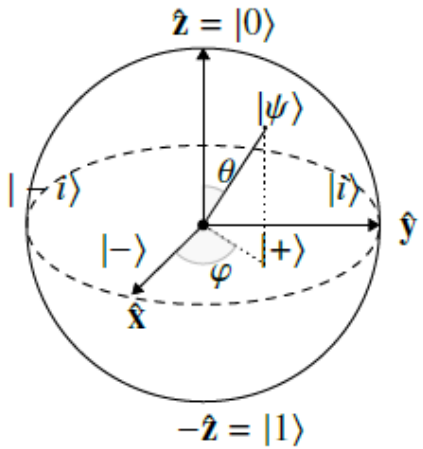
If $\theta = \pi \implies |1\rangle$

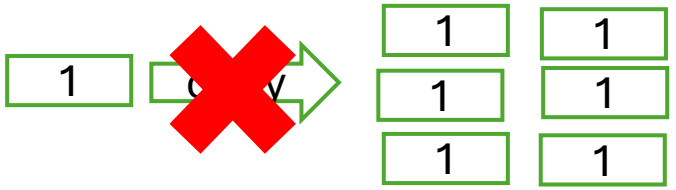
If $\theta = \frac{\pi}{2} \wedge \phi = 0 \implies |+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$

If $\theta = \frac{\pi}{2} \wedge \phi = \pi \implies |-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$

If $\theta = \frac{\pi}{2}, \phi = \frac{\pi}{2} \implies |i\rangle = \frac{|0\rangle+i|1\rangle}{\sqrt{2}}$

If $\theta = \frac{\pi}{2}, \phi = \frac{3\pi}{2} \implies |-i\rangle = \frac{|0\rangle-i|1\rangle}{\sqrt{2}}$





Quantum gates

- **Matrix form**
- **Unitary:** $U^{\dagger}U = I$
- **Reversible :** $U^{-1}U = I$
- **Linearity:** $U(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha U(\psi\rangle) + \beta U(|\phi\rangle)$

$\mathbb{I}$	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
S	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$
$S^{\dagger}$	$\begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$
$T^{\dagger}$	$\begin{pmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix}$

$$R_x(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

$$R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$$

$$CX = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$SWAP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$CSWAP = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$CCX = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Quantum gates and entanglement

Entanglement is a typical quantum phenomenon for which the state of two or more qubits cannot be written anymore as a combination of separated qubits

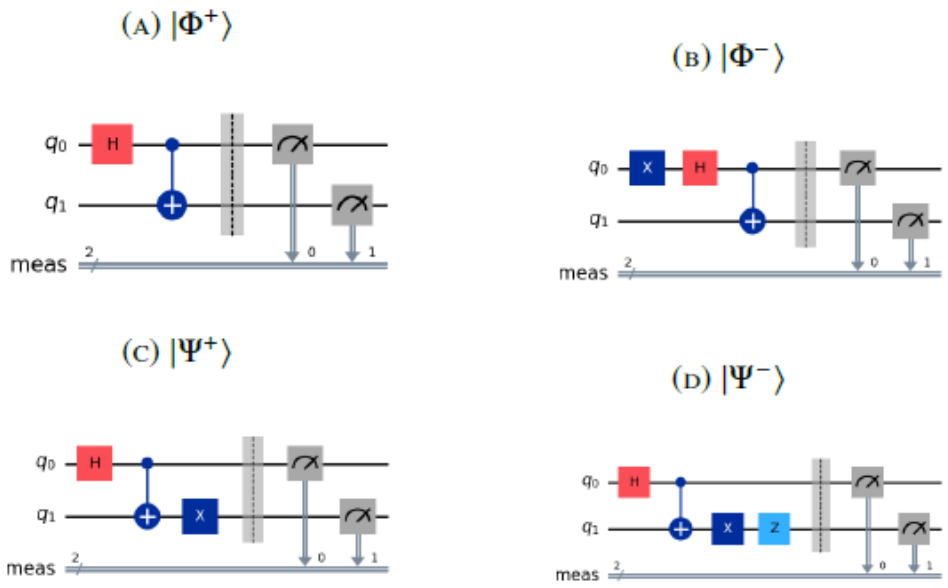
Certain specific combinations generate entanglement

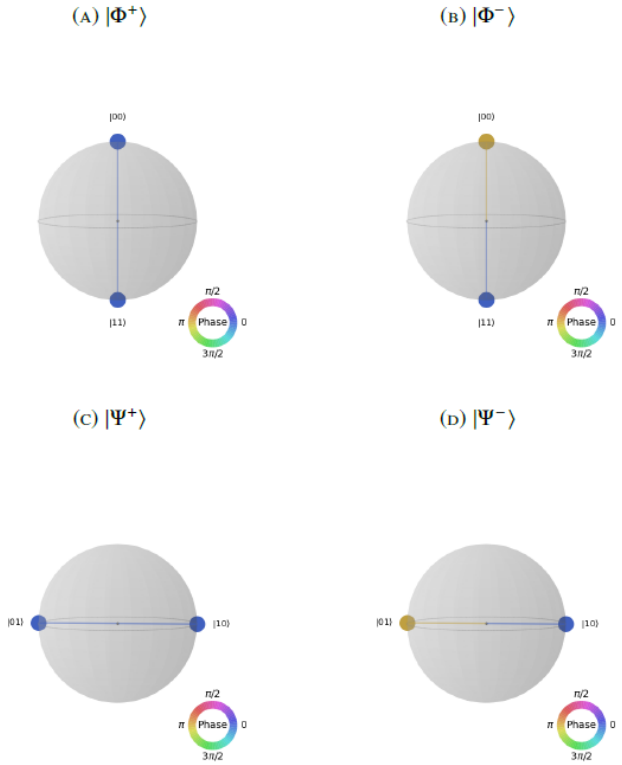
$$|\Phi^+\rangle = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$$

$$|\Phi^-\rangle = \frac{|00\rangle-|11\rangle}{\sqrt{2}}$$

$$|\Psi^+\rangle = \frac{|01\rangle+|10\rangle}{\sqrt{2}}$$

$$|\Psi^-\rangle = \frac{|01\rangle-|10\rangle}{\sqrt{2}}$$





Quantum Support Vector Machines
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Theoretical intuition

Why Quantum Support Vector Machines

- Classical SVM performance depends on the choice of kernel and its ability to capture non-linear feature interactions (Cortes & Vapnik, 1995; Vapnik, 1998)

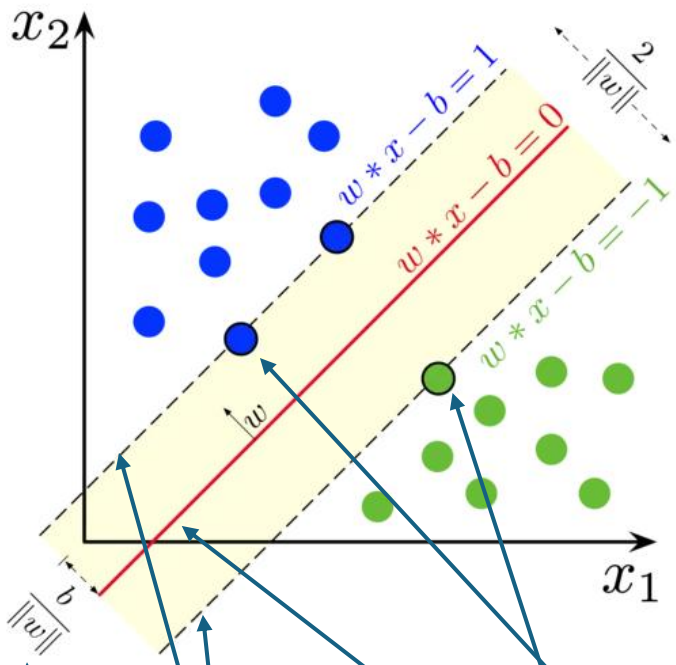
- Increasing kernel complexity improves expressivity but raises computational and scalability issues (Schölkopf & Smola, 2002)

- Quantum computing enables implicit data embedding into high-dimensional Hilbert spaces through quantum feature maps (Schuld & Killoran, 2019)

- Quantum kernels exploit superposition and entanglement to encode complex, global feature correlations (Havlíček et al., 2019)

- Enhanced feature expressivity may improve class separability under strong regularization regimes (Schuld et al., 2021)

Classical Support Vector Machines



Support vectors

Decision boundary

Maximum Margin

Positive and negative hyperplanes

$\vec{w} \cdot \vec{x}_i - b \geq +1$ if $y_i = +1$

$\vec{w} \cdot \vec{x}_i - b \leq -1$ if $y_i = -1$

$$\max \frac{2}{\|w\|} \quad \text{s.t. } y_i(\mathbf{w}^T x_i + b) \geq 1 \ \forall i$$

Primal formulation $\min \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N} \xi_i \quad \text{s.t. } y_i(\mathbf{w}^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \ \forall i$

Dual formulation $L(\mathbf{w}, b, \xi, \alpha, \mu) = \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N} \xi_i - \sum_{i=1}^{N} \alpha_i[y_i(\mathbf{w}^T x_i + b) - 1 + \xi_i] - \sum_{i=1}^{N} \mu_i \xi_i$

KKT conditions

$\frac{\partial L}{\partial \mathbf{w}} = 0 \implies \mathbf{w} = \sum_{i=1}^{N} \alpha_i y_i x_i$

$\frac{\partial L}{\partial b} = 0 \implies \sum_{i=1}^{N} \alpha_i y_i = 0$

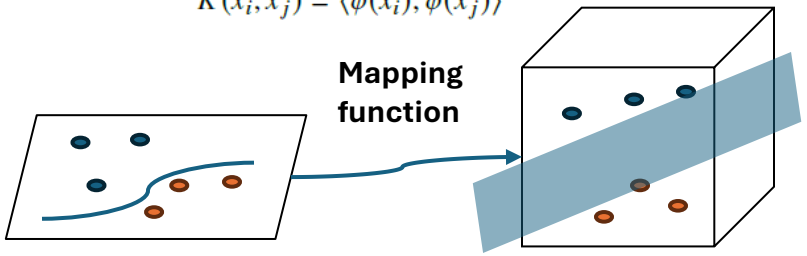
$\frac{\partial L}{\partial \xi_i} = 0 \implies \alpha_i + \mu_i = C$

$$\max_\alpha \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\sum_{i,j=1}^{N} \alpha_i \alpha_j y_i y_j (x_i^T x_j) \quad \text{s.t. } 0 \leq \alpha_i \leq C, \sum_{i=1}^{N} \alpha_i y_i = 0$$

$$\max_\alpha \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\sum_{i,j=1}^{N} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \quad \text{s.t. } 0 \leq \alpha_i \leq C, \sum \alpha_i y_i = 0$$

$$K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle$$

Mapping function



$$f(x) = \sum_{i=1}^{N} \alpha_i y_i K(x_i, x) + b$$

Linear: $K(x_i, x_j) = x_i^T x_j$

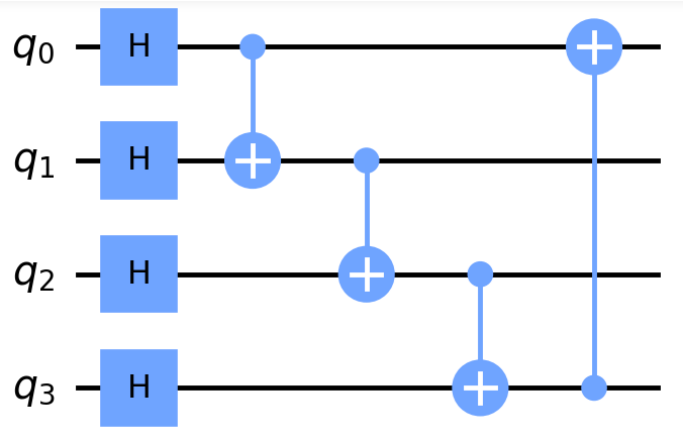
Polynomial: $K(x_i, x_j) = (x_i^T x_j + c)^d$

Radial Basis Function: $K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2}$

Quantum Support Vector Machines

Features are transformed in quantum states (must be normalized to the interval (0,1) to be readable by the quantum feature map)

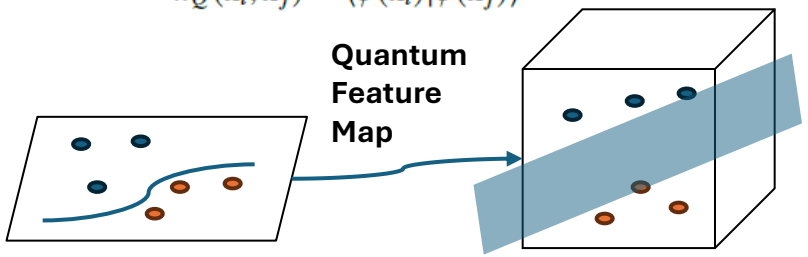
$$\mathbf{x} \mapsto |\phi(\mathbf{x})\rangle = U_\phi(\mathbf{x})\,|0\rangle$$



Quantum Feature Map

$$\max_\alpha \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j k_Q(\mathbf{x}_i, \mathbf{x}_j) \quad \text{s.t. } 0 \le \alpha_i \le C, \sum_{i=1}^{N} \alpha_i y_i = 0$$

$$k_Q(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i) | \phi(\mathbf{x}_j) \rangle$$



$$f(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i^* y_i k_Q(\mathbf{x}_i, \mathbf{x}) + b^*$$

Quantum Support Vector Machines
-
Multiclass Classification case study

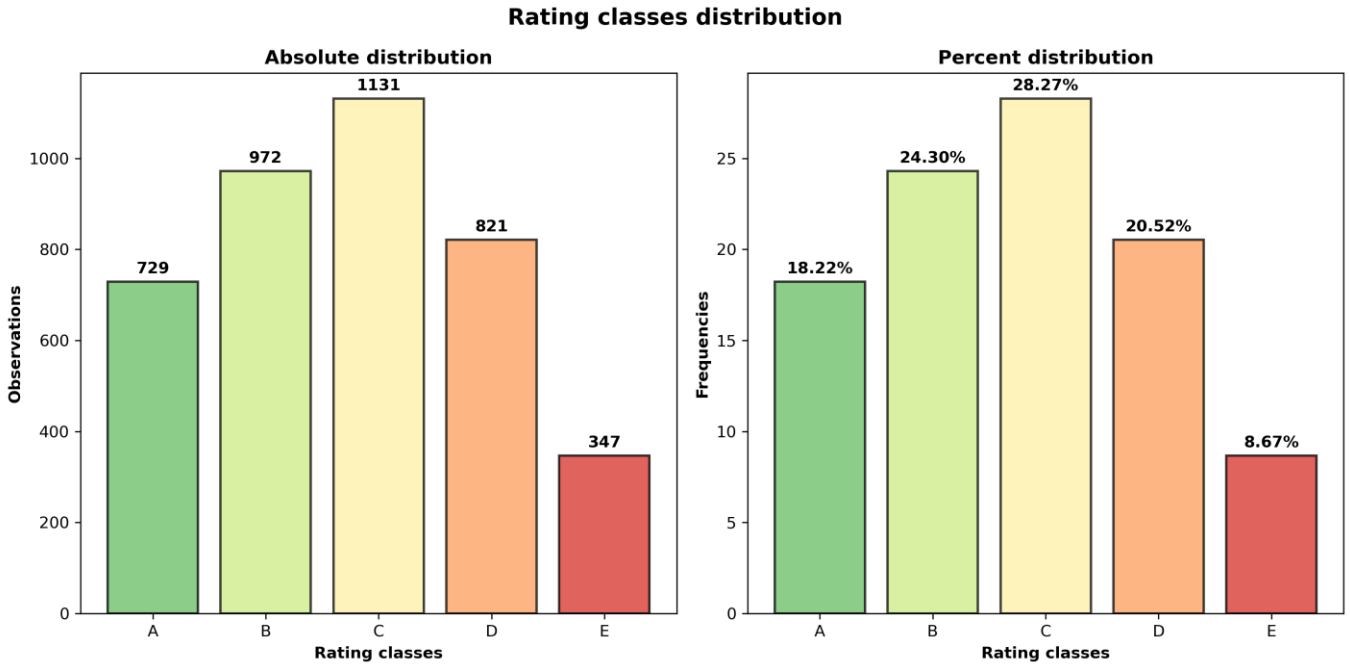
Dataset analysis

4000 anonymous US publicly traded manufacturing firms , divided into 5 credit score classes (from A to E) on the basis of their own Altman Z-score

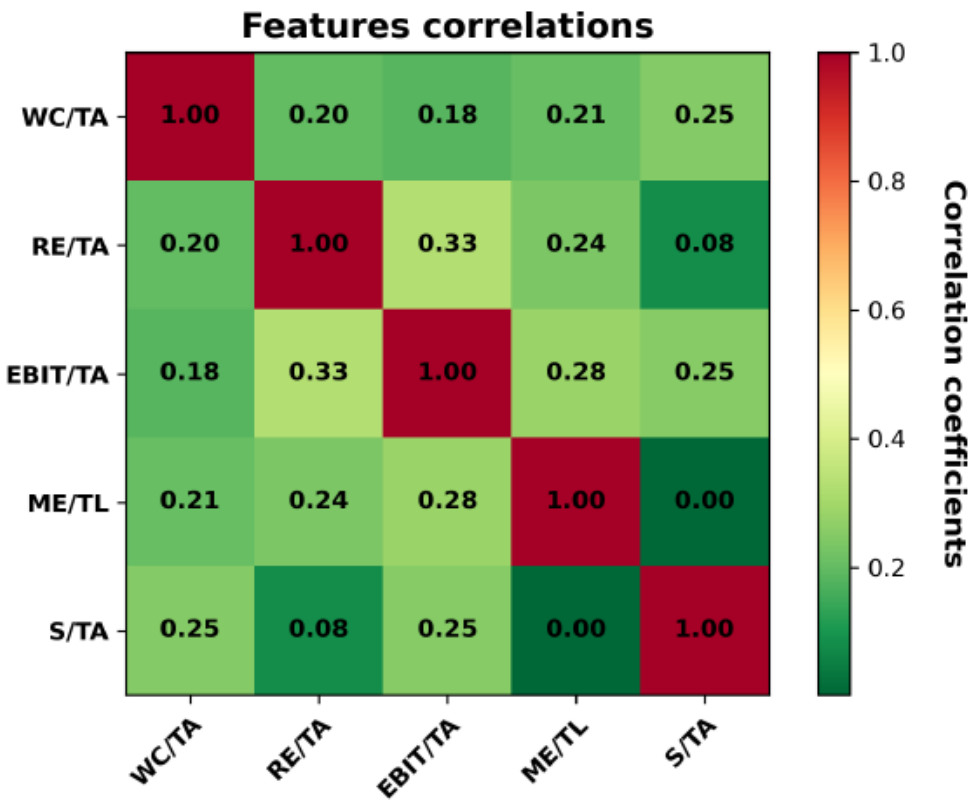
$$Z = 1.2\frac{WC}{TA} + 1.4\frac{RE}{TA} + 3.3\frac{EBIT}{TA} + 0.6\frac{ME}{TL} + 1.0\frac{S}{TA}$$

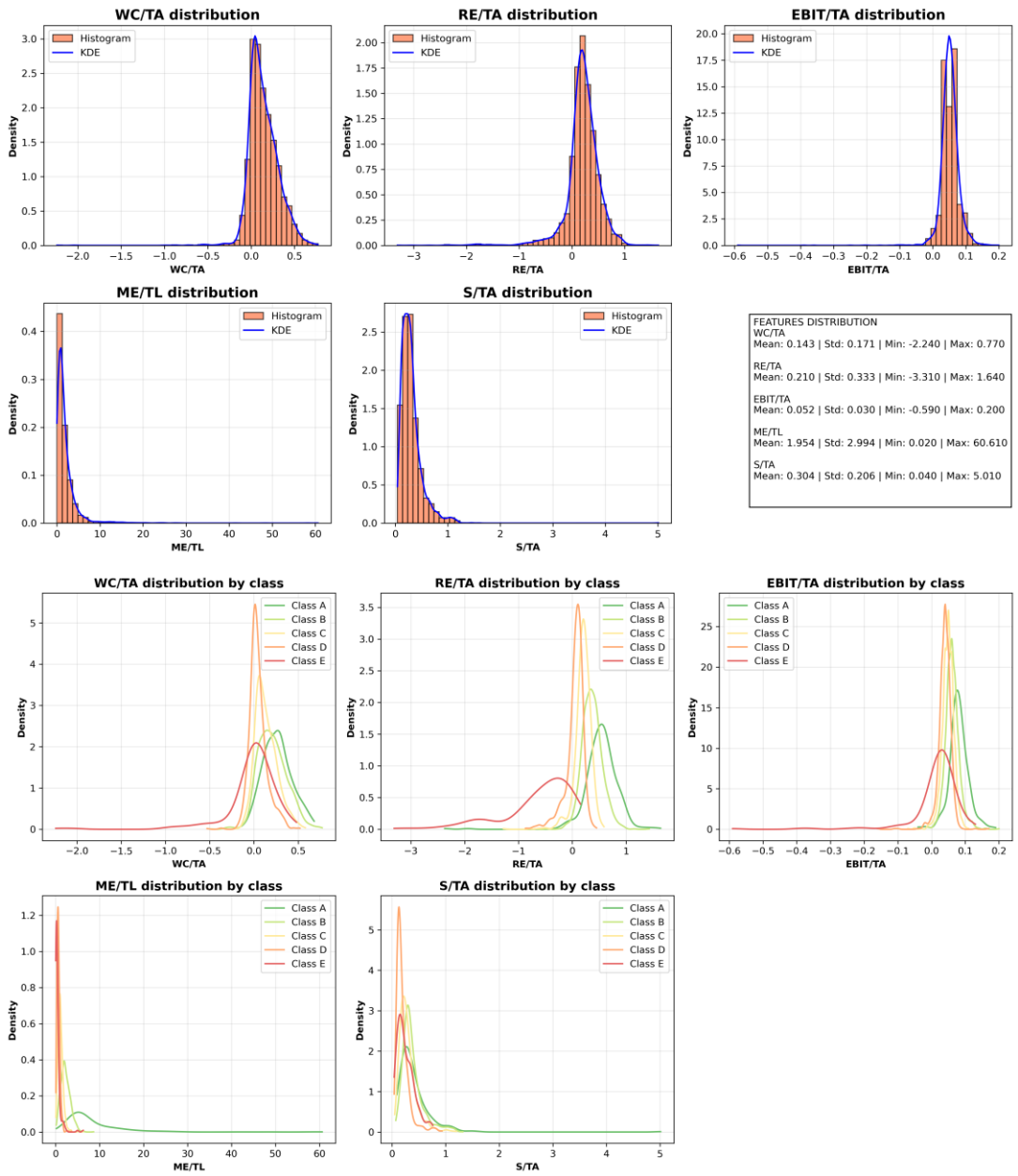
- **WC/TA = Working capital/Total assets (financial solidity)**
- **RE/TA = Retained earnings/Total assets (maturity and stability indicator)**
- **EBIT/TA = Earnings before interest and tax/Total assets (operative profitability)**
- **ME/TL = Market Value of Equity / Book Value of Total Liabilities (financial leverage indicator)**
- **S/TA = Sales/Total assets (managerial efficiency)**

$$Rating = \begin{cases} A & \text{if } Z > 3.25 \\ B & \text{if } 1.65 < Z \leq 3.25 \\ C & \text{if } 1 < Z \leq 1.65 \\ D & \text{if } 0.25 < Z \leq 1 \\ E & \text{if } Z \leq 0.25 \end{cases}$$



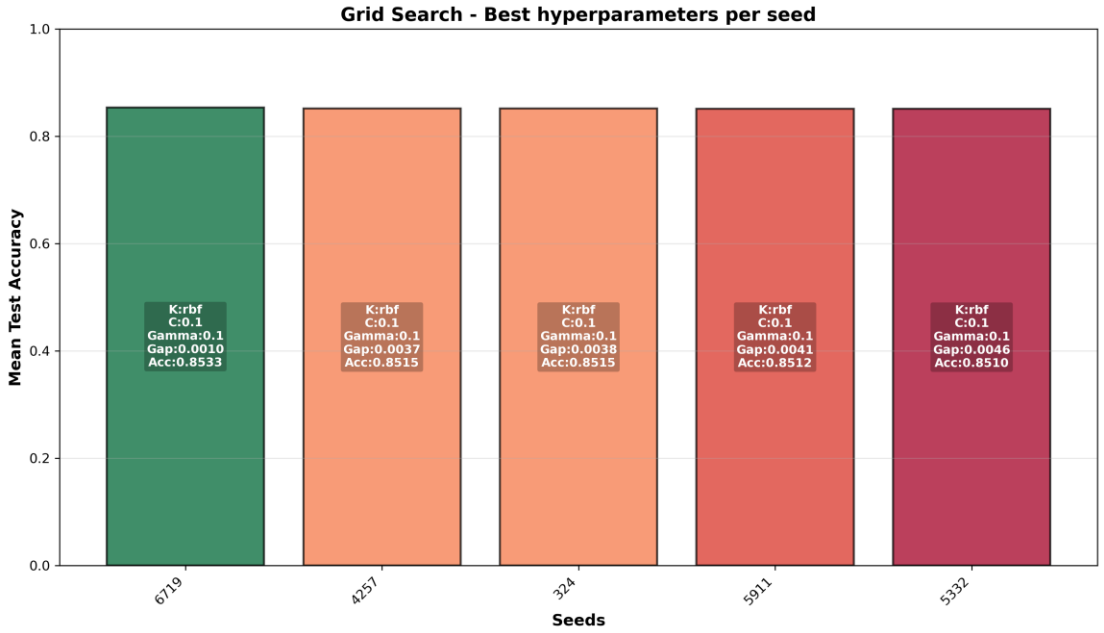
Dataset analysis

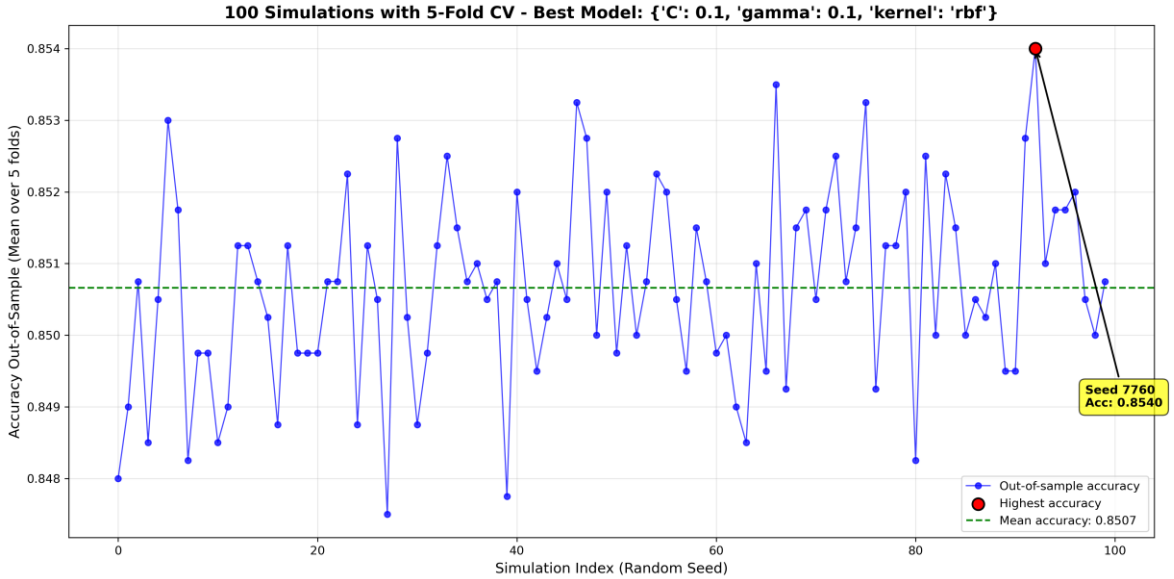




Two-step benchmark model selection

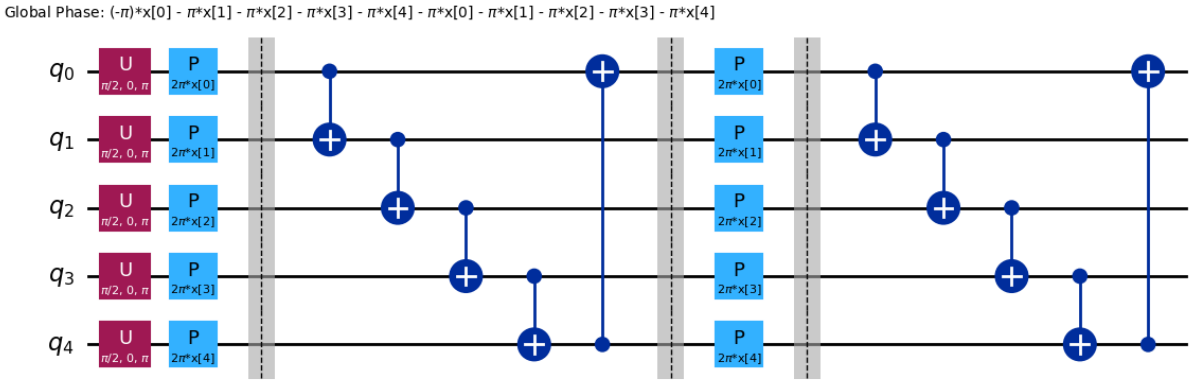
- **Grid search combinations**
  - **Kernel: Radial Basis Function** $e^{-\gamma\|x-y\|^2}$
  - $C \in \{0.1, 1, 10, 25\}$
  - $\gamma \in \{0.001, 0.01, 0.1, 1\}$
- **100 simulations with the best hyperparameters configuration**
- **5-fold CV in all the steps**





Implementation of One-Vs-One technique (sum of all binary classification among all the combinations of classes)

Quantum feature maps

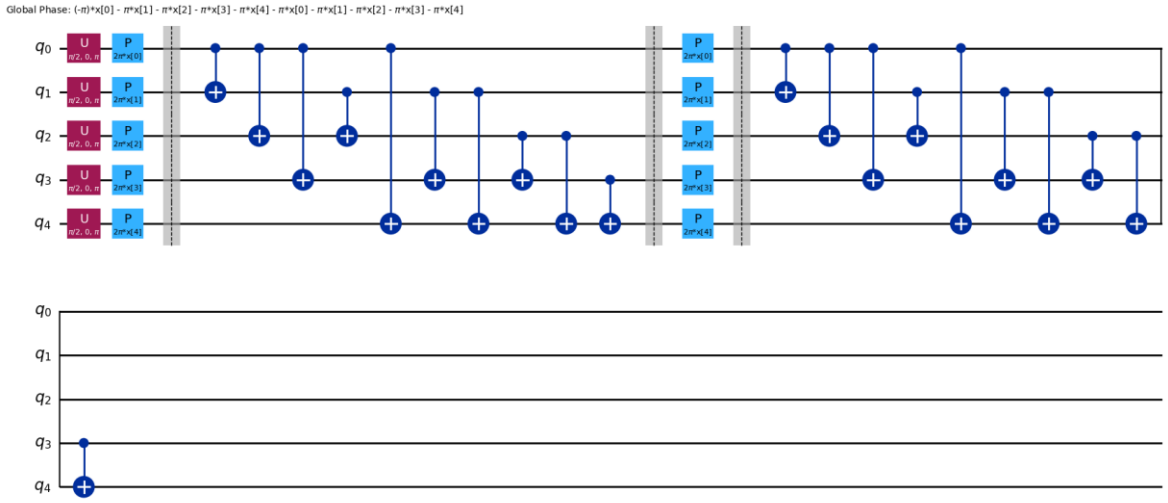


Circularly entangled R_z gates

- Parametrized Z-rotation gate to each adjacent x_i
- Locally limited entanglement → less complexity, better stability
- Ideal for near correlations

Fully entangled R_z gates

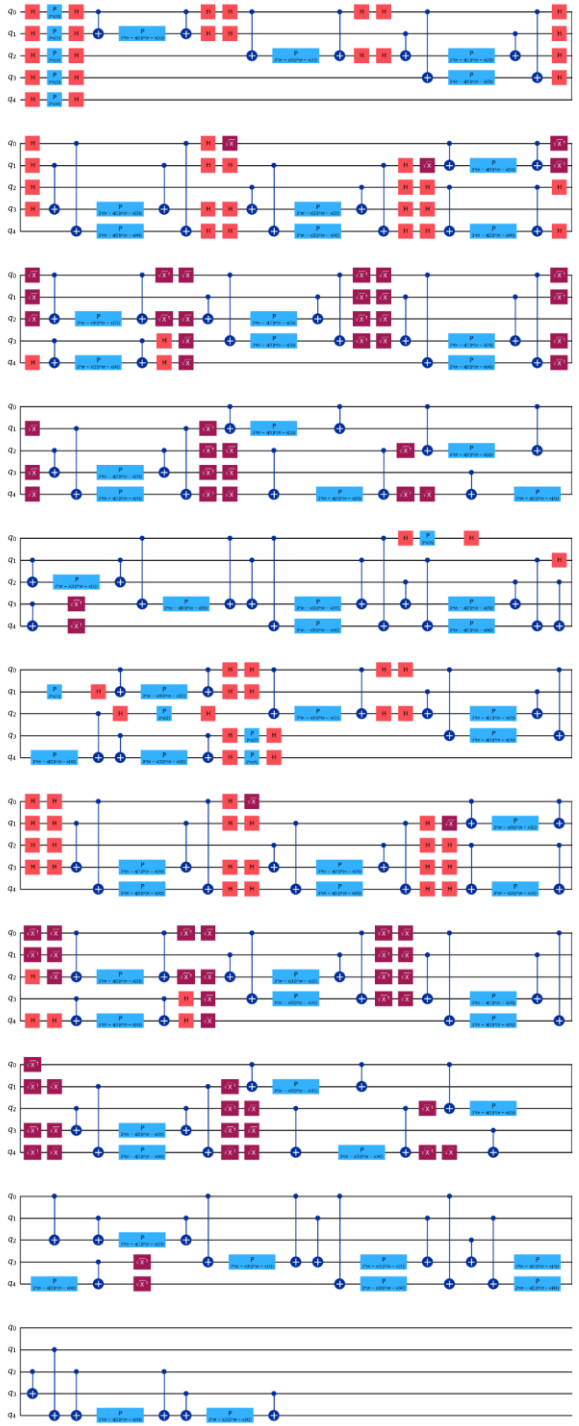
- Parametrized Z-rotation gate to each x_i
- Full entanglement → higher complexity
- Ideal for capturing non-linear global correlations



Quantum feature maps

Fully entangled complete Pauli

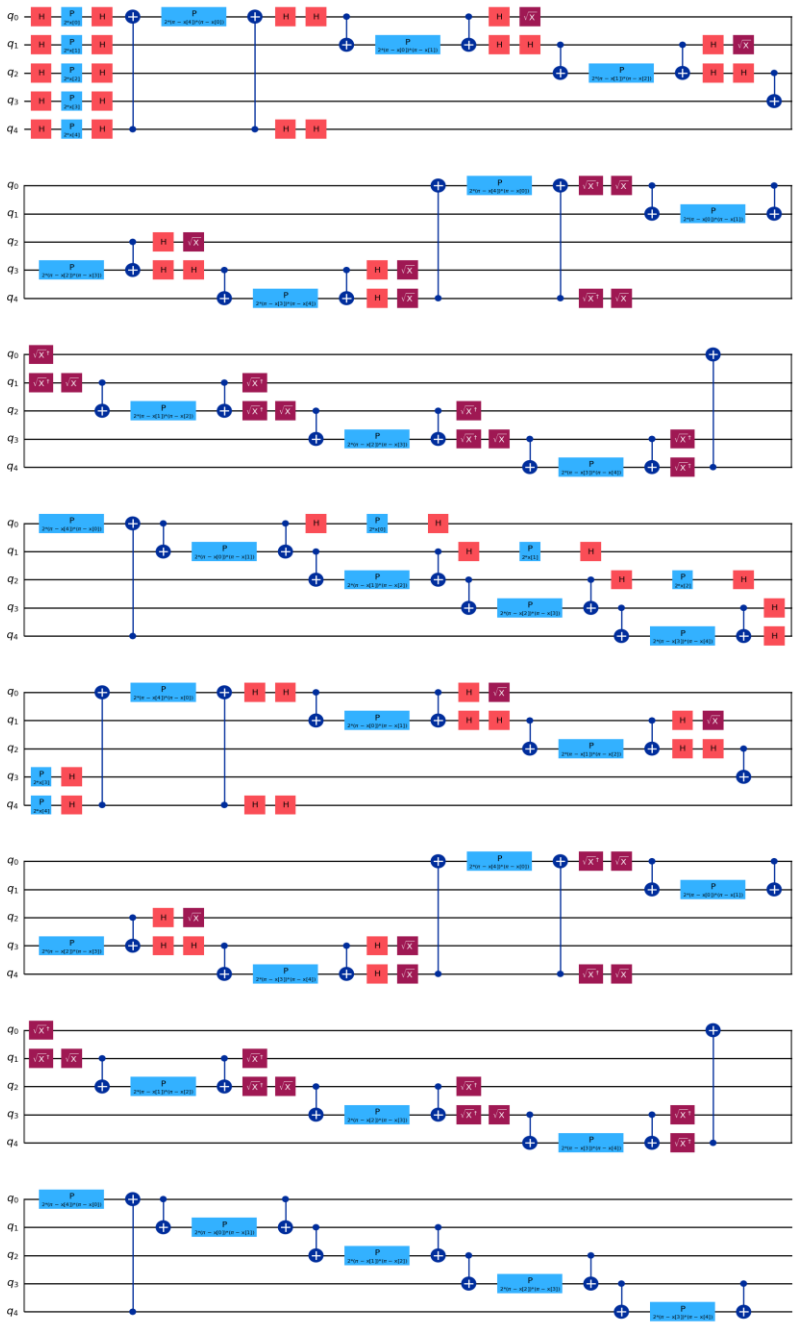
- Parametrized rotations using Pauli gates X, Y, Z
- Applied to each qubit
- Multilinear interactions among features through combinations $X_i X_j, Y_i Y_j, Z_i Z_j$
- Highly expressive structure
- Repeated entanglement among all features
- Optimal to model high-dimensionality non-linear correlations
- Potentially more powerful than previous feature maps but higher computational cost



Quantum feature maps

Circularly entangled complete Pauli

- Similar to the previous complete Pauli feature map with the only difference in entanglement application
- Each qubit is entangled with the adjacent and the last with the first
- Intermediate complexity and regular connection pattern
- Multilinear interactions among neighbouring features
- Less noisy than fully entangled version

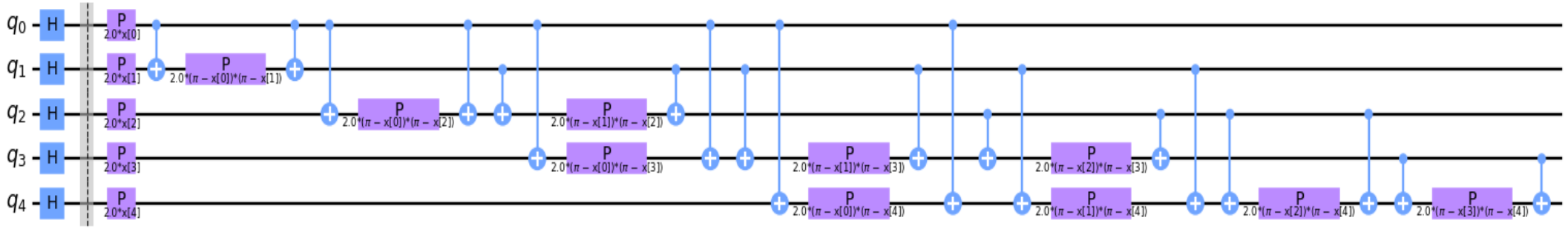


Quantum feature maps

ZZFeatureMaps

Fully entangled ZZFeatureMap/ZZ Pauli feature maps

- R_z and ZZ gates applied to all single qubits
- Global quadratic correlations for each feature couple (x_i, x_j)
- Deeper and more complex structure with $O(n^2)$ interactions
- Higher computational cost
- Ideal to detect global correlations



Final outcomes

Fully entangled ZZFeatureMap

- Best performance (0.928, 0.929)
- Captures global correlations among all the features
- Extremely stable results (low accuracy gap)
- Particularly efficient in classifying central classes

Fully entangled Pauli

- Quite lower performance (0.91)
- Sensitivity to the circuit complexity (higher accuracy gap)
- Good compromise to capture mulitlinear global interactions
- Optimal performance in the classification of classes B and C, quite lower in extreme ones

Fully entangled Rz

- Solid outcomes, close to the fully entangled Pauli (0.909)
- Good balance and structured kernels
- Better generalization with respect to the complete Pauli (lower gap)
- Higher performance than circularly entangled version

Classical SVM

- Worst performance (0.854)
- QSVMs generate quantum kernel matrices with higher expressivity
- Lower capacity to capture central adjacent classes

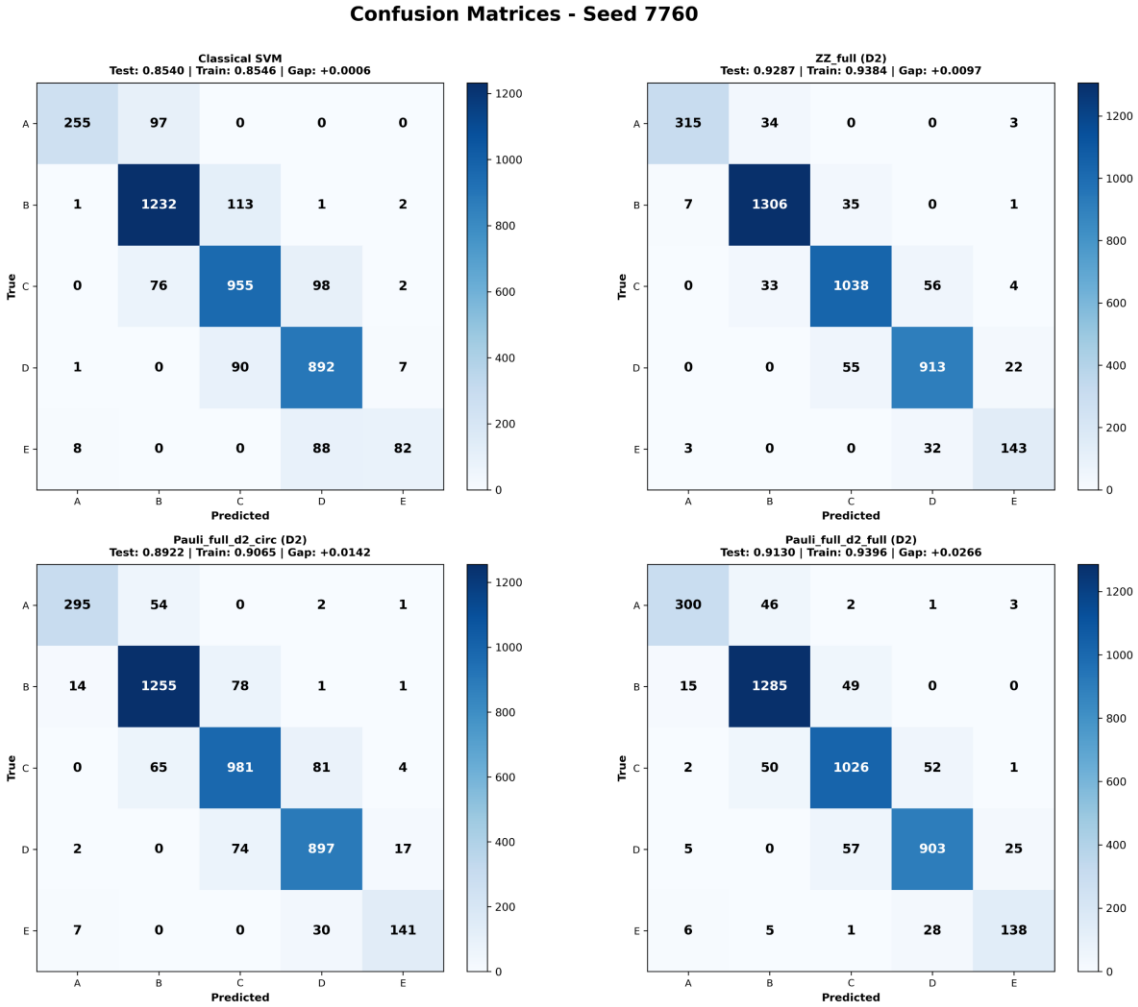
Circularly entangled Pauli

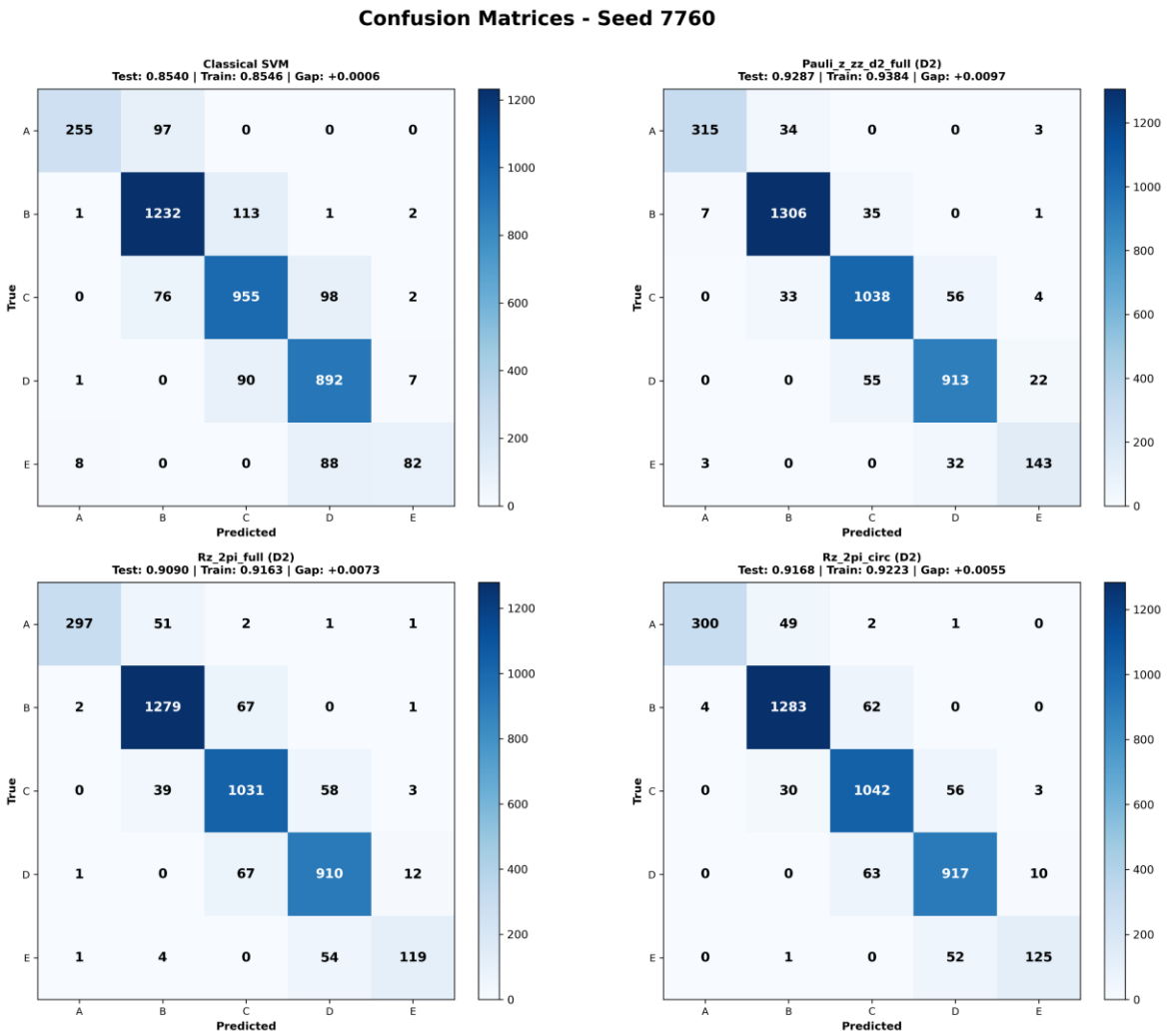
- Lower performance than fully entangled version (0.892)
- Local entanglement
- Extremely stable results (low accuracy gap)
- Particularly efficient in classifying central classes

Circularly entangled Rz

- Good performances (0.916)
- Structural limits due to the local entanglement
- Good stability for faster simulations

Final outcomes





Final considerations

- The analysis confirms the potential of the quantum kernel method as a competitive approach in multiclass problems with a strong structure in the features.
- Quantum-enhanced models overperform classical ones for their capacity to capture more complex non-linear correlations between variables thanks to the creation of entanglement phenomena.
- Advantage in credit emission evaluation process
- Lower credit loss for financial and insurance institutions.
- The main limitations remain circuit depth and computational cost in the more expressive models, other than their high costs of in-house quantum computing infrastructure implementation.

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Thank you for your attention!

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